Integration & Applications

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Area under a curve**

Find the area under $y = x^2$ from $x = 0$ to $x = 3$, using a definite integral.

Area $= \int_0^3 x^2\,dx = \left[\frac{x^3}{3}\right]_0^3 = \frac{27}{3} = 9.$

2 **[Foundational]** • **Target: Area under a curve**

Find the area under $y = 4 - x^2$ from $x = -2$ to $x = 2$.

$F(x) = 4x - \frac{x^3}{3}$. $F(2) = 8 - \frac{8}{3} = \frac{16}{3}$. $F(-2) = -8 + \frac{8}{3} = -\frac{16}{3}$. Area $= \frac{16}{3} - \left(-\frac{16}{3}\right) = \frac{32}{3}$.

3 **[Foundational]** • **Target: Signed area and regions below the axis**

Evaluate $\int_0^2 (x - 2)\,dx$ and explain what the sign of the result means.

$\left[\frac{x^2}{2} - 2x\right]_0^2 = (2 - 4) - 0 = -2$. The negative sign means the curve lies below the x-axis on $[0, 2]$.

4 **[Foundational]** • **Target: Signed area and regions below the axis**

Find the (positive) area between $y = x - 2$ and the x-axis, from $x = 0$ to $x = 2$.

The signed integral is -2 (from the previous question), so the area is $|-2| = 2$.

5 **[Foundational]** • **Target: Finding a function from its derivative**

A function satisfies $f'(x) = 6x^2 - 4$ and $f(1) = 5$. Find $f(x)$.

$f(x) = 2x^3 - 4x + C$. $f(1) = 2 - 4 + C = 5 \Rightarrow C = 7$. $f(x) = 2x^3 - 4x + 7$.

6 **[Foundational]** • **Target: Average value of a function**

Find the average value of $f(x) = x^2$ on $[0, 3]$, using $\text{avg} = \frac{1}{b-a}\int_a^b f(x)\,dx$.

$\text{avg} = \frac{1}{3}\int_0^3 x^2\,dx = \frac{1}{3}(9) = 3.$

7 **[Foundational]** • **Target: Average value of a function**

Find the average value of $f(x) = 2x + 1$ on $[1, 5]$.

$\int_1^5 (2x + 1)\,dx = [x^2 + x]_1^5 = 30 - 2 = 28$. $\text{avg} = \frac{28}{5-1} = 7$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Word problems: integration in context

This question has two parts. A particle moves along a line with velocity $v(t) = 6t - 3t^2$ (m/s) for $0 \le t \le 4$.

a Find the particle's displacement from $t = 0$ to $t = 2$, using $\int_0^2 v(t)\,dt$.

$\int_0^2 (6t - 3t^2)\,dt = [3t^2 - t^3]_0^2 = 12 - 8 = 4$ meters.

b Find the particle's displacement from $t = 0$ to $t = 4$, and explain why it is not simply twice the part (a) answer.

$\int_0^4 (6t - 3t^2)\,dt = [3t^2 - t^3]_0^4 = 48 - 64 = -16$ meters. The velocity becomes negative for $t > 2$ (the particle reverses direction), so the displacements over $[0, 2]$ and $[2, 4]$ partially cancel rather than simply add.

(a) $\int_0^2 (6t - 3t^2)\,dt = [3t^2 - t^3]_0^2 = 12 - 8 = 4$ meters.; (b) $\int_0^4 (6t - 3t^2)\,dt = [3t^2 - t^3]_0^4 = 48 - 64 = -16$ meters. The velocity becomes negative for $t > 2$ (the particle reverses direction), so the displacements over $[0, 2]$ and $[2, 4]$ partially cancel rather than simply add.

9 [Intermediate] • Target: Word problems: integration in context

This question has two parts. Water flows into a tank at a rate of $r(t) = 20 - 2t$ liters per minute, for $0 \le t \le 10$.

a Write a definite integral for the total volume of water that flows in during the first 5 minutes.

$V = \int_0^5 (20 - 2t)\,dt$.

b Evaluate the integral to find the volume.

$V = [20t - t^2]_0^5 = 100 - 25 = 75$ liters.

(a) $V = \int_0^5 (20 - 2t)\,dt$.; (b) $V = [20t - t^2]_0^5 = 100 - 25 = 75$ liters.

10 [Intermediate] • Target: Finding a function from its derivative

Find the area under the line $y = x$ above the x-axis from $x = 0$ to $x = 4$.

Step 1: Set up the definite integral: The area under a continuous non-negative function $y = f(x)$ from a to b is given by: $A = \int_0^4 x\,dx$ Step 2: Evaluate via Fundamental Theorem of Calculus: $A = \left[\frac{x^2}{2}\right]_0^4 = \frac{4^2}{2} - \frac{0^2}{2} = \frac{16}{2} - 0 = \mathbf{8}$ Geometric verification: The region is a right triangle with base 4 and height 4: Area $= \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 4 \times 4 = 8$ ✓

11 [Intermediate] • Target: Average value of a function

Evaluate the definite integral: $\int_0^3 (2x + 1)\,dx$

Step 1: Find the antiderivative: $\int (2x + 1)\,dx = x^2 + x$. Step 2: Evaluate at the limits 0 and 3: $[x^2 + x]_0^3 = (3^2 + 3) - (0^2 + 0) = (9 + 3) - 0 = \mathbf{12}$.

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12 **[Intermediate]** • **Target: Word problems: integration in context**

The function $f(x) = x^2 - 6x + k$ represents a parabola that touches the x-axis at $x = 3$. What is the value of k?

For the parabola to touch the x-axis at $x = 3$, $x = 3$ must be the only root of the equation $f(x) = 0$. This implies that $f(3) = 0$ and the discriminant $\Delta = b^2 - 4ac$ must be zero. Substituting $x = 3$ into the function: $f(3) = (3)^2 - 6(3) + k = 0$ $9 - 18 + k = 0$ $-9 + k = 0$ $k = 9$ Alternatively, we can use the fact that $x = 3$ is the x-coordinate of the vertex of the parabola. The vertex coordinates (h, k) for the function $ax^2 + bx + c$ are $h = -\frac{b}{2a}$. In this case, $a = 1$ and $b = -6$..

13 **[Intermediate]** • **Target: MCQ — Basic antiderivatives (the power rule)**

The graph of the function $f(x)$ is shown below. What is the value of $\lim_{x \to 2} f(x)$?

To find the value of the limit of the function $f(x)$ as x approaches 2 ($\lim_{x \to 2} f(x)$) from the graph, we look for the y value that the function approaches as we get closer to $x = 2$ from both sides (left and right). Looking at the graph, as x approaches 2 from the left (values less than 2), $f(x)$ approaches 2. As x approaches 2 from the right (values greater than 2), $f(x)$ also approaches 2. Since the limit from the left equals the limit from the right, the limit exists and its value is.

14 **[Intermediate]** • **Target: MCQ — Antiderivatives of polynomials**

In the diagram below, the straight line $y = mx + c$ intersects the x and y axes. If its slope is $\frac{1}{2}$ and it passes through the point $(4, 5)$, find the value of the constant term c.

The general equation of a straight line is $y = mx + c$, where m is the slope and c is the constant term (y-intercept). We know that the slope $m = \frac{1}{2}$ and the line passes through the point $(4, 5)$. We can substitute these values into the equation to find c: $5 = \left(\frac{1}{2}\right)(4) + c$ $5 = 2 + c$ Subtracting 2 from both sides: $c = 5 - 2$ $c = 3$

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: MCQ — Antiderivatives of trigonometric and exponential functions** • **Cross-Lesson: 4.01 Trigonometry**

If $\int_1^4 f(x)\,dx = 7$ and $\int_2^4 f(x)\,dx = 3$, determine the value of $\int_1^2 f(x)\,dx$.

Step 1: Apply the interval additivity property: $\int_1^4 f(x)\,dx = \int_1^2 f(x)\,dx + \int_2^4 f(x)\,dx$. Step 2: Substitute given values and solve: $7 = \int_1^2 f(x)\,dx + 3 \Rightarrow \int_1^2 f(x)\,dx = 7 - 3 = \mathbf{4}$.

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16 [Challenging] • Target: MCQ — Evaluating definite integrals • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Find the general indefinite integral: $\int (4x^3 - 6x + 5)\,dx$

Step 1: Integrate term-by-term using the power rule: $\int 4x^3\,dx = x^4$, $\int -6x\,dx = -3x^2$, $\int 5\,dx = 5x$. Step 2: Append the constant of integration C: $\mathbf{x^4 - 3x^2 + 5x + C}$.

17 [Challenging] • Target: MCQ — Definite integrals with polynomial integrands • Cross-Lesson: 6.01 Limits & Derivatives

The function $f(x) = x^2 - 6x + 9$ represents the trajectory of a projectile. What is the minimum height the object reaches?

The function $f(x) = x^2 - 6x + 9$ is a quadratic function in the form $ax^2 + bx + c$ where $a = 1$, $b = -6$, and $c = 9$. Since $a > 0$, the parabola opens upwards, meaning the function has a minimum value. The x -coordinate of the vertex can be found using the formula $x = -b/(2a)$. $x = -(-6)/(2 * 1) = 6/2 = 3$ To find the minimum height (the y-value of the vertex), we substitute $x = 3$ into the original function: $f(3) = (3)^2 - 6(3) + 9 = 9 - 18 + 9 = 0$ Therefore, the minimum height the object rea.

18 [Challenging] • Target: MCQ — Definite integrals of trigonometric and exponential functions • Cross-Lesson: 8.01 Exponentials & Logarithms

Level 3 · Advanced Find the exact surface area generated by revolving the line segment $y = 2x$ on $[0, 3]$ about the x-axis.

Here $f(x) = 2x$ and $f'(x) = 2$. The arc length element is $ds = \sqrt{1 + (f'(x))^2}\,dx = \sqrt{1 + 2^2}\,dx = \sqrt{5}\,dx$. The surface area formula gives: $S = 2\pi\int_0^3 f(x)\sqrt{1 + [f'(x)]^2}\,dx = 2\pi\int_0^3 (2x)\sqrt{5}\,dx = 4\sqrt{5}\pi\left[\frac{x^2}{2}\right]_0^3 = 4\sqrt{5}\pi\left(\frac{9}{2}\right) = 18\sqrt{5}\pi$ Verification via cone geometry: The shape is a right circular cone with radius $r = f(3) = 6$ and slant height $l = \sqrt{3^2 + 6^2} = \sqrt{45} = 3\sqrt{5}$. Lateral are.

19 [Challenging] • Target: MCQ — Area under a curve • Cross-Lesson: 9.01 Statistics & Probability

The graph of the function $f(x)$ is shown. What is the limit of $f(x)$ as x approaches 2?

We look for the value that $f(x)$ approaches as x approaches 2 from both sides. Looking at the graph, as x approaches 2 from the left (values less than 2), $f(x)$ approaches 2. As x approaches 2 from the right (values greater than 2), $f(x)$ also approaches 2. Since the left-hand limit equals the right-hand limit, the limit of $f(x)$ as x approaches 2 is 2.

20 [Challenging] • Target: MCQ — Signed area and regions below the axis • Cross-Lesson: 1.01 Algebra & Equations

A particle is projected vertically upwards. The function $f(t) = -5t^2 + 20t + 1$ represents the height (in meters) of the particle after t seconds. What is the maximum height reached by the particle?

The given function is a quadratic function, $f(t) = -5t^2 + 20t + 1$, which represents a downward-opening parabola because the coefficient of t^2 is negative (-5). The maximum height of the particle corresponds to the maximum value of the quadratic function, which occurs at the vertex of the parabola. The vertex coordinates (h, k) of a quadratic function $at^2 + bt + c$ are given by $h = -\frac{b}{2a}$. In this function, $a = -5$ and $b = 20$. The time to reach maximum height is.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Area under a curve

The graph shows $y = x^2$. Find the area under the curve from $x = 0$ to $x = 3$, using a definite integral.

a. 9 ✓

b. 18

c. 3

d. 27

$\left[\frac{x^3}{3}\right]_0^3 = 9$.

22 [Foundational] • Target: MCQ — Area under a curve

The graph shows $y = 4 - x^2$. Find the area under the curve from $x = -2$ to $x = 2$.

a. $\frac{16}{3}$

b. 32

c. 16

d. $\frac{32}{3}$ ✓

$\left[4x - \frac{x^3}{3}\right]_{-2}^2 = \frac{16}{3} - \left(-\frac{16}{3}\right) = \frac{32}{3}$.

23 [Foundational] • Target: MCQ — Signed area and regions below the axis

Evaluate $\int_0^4 (x - 4)\,dx$ and state whether the region is above or below the x-axis.

a. 8; above

b. -8; above

c. 8; below

d. -8**; below** ✓

$\left[\frac{x^2}{2} - 4x\right]_0^4 = 8 - 16 = -8$; negative means below the x-axis.

24 [Foundational] • Target: MCQ — Word problems: integration in context

A particle moves along a line with velocity $v(t) = 8t - 2t^2$ (m/s) for $0 \le t \le 4$. Find the particle's displacement over this interval.

a. 64

b. $\frac{64}{3}$ ✓

c. 32

d. $\frac{192}{3}$

$\int_0^4 (8t - 2t^2)\,dt = \left[4t^2 - \frac{2t^3}{3}\right]_0^4 = \frac{64}{3}$ m.

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25 [Foundational] • Target: MCQ — Word problems: integration in context

Water flows into a tank at a rate of $r(t) = 30 - 3t$ liters per minute, for $0 \le t \le 10$. Find the total volume of water that flows in during this time.

a. 300

b. 150 ✓

c. 450

d. 100

$\int_0^{10} (30 - 3t)\,dt = [30t - 1.5t^2]_0^{10} = 150$ liters.

26 [Foundational] • Target: MCQ — Word problems: integration in context

The graph shows a particle's velocity $v(t) = 8t - 2t^2$ for $0 \le t \le 4$. Find the particle's displacement (the area under the curve).

a. $\frac{64}{3}$ ✓

b. 16

c. 32

d. 64

$\int_0^4 (8t - 2t^2)\,dt = \frac{64}{3}$ m.

27 [Foundational] • Target: MCQ — Word problems: integration in context

A factory's production rate is $p(t) = 50 + 4t$ items per hour, for $0 \le t \le 8$. Find the total number of items produced during this 8-hour period.

a. 528 ✓

b. 656

c. 450

d. 400

$\int_0^8 (50 + 4t)\,dt = [50t + 2t^2]_0^8 = 528$ items.

28 [Foundational] • Target: Signed area and regions below the axis

[redacted] $f(x) = 2x$ [redacted] $x = 1$ [redacted] $x = 3$[redacted]. To find the area under the curve of the function $f(x) = 2x$ from $x = 1$ to $x = 3$, we calculate the definite integral.

a. 4

b. 6

c. **8** ✓

d. 10

[redacted] $f(x) = 2x$ [redacted] $x = 1$ [redacted] $x = 3$. $\int_1^3 2x\,dx = [x^2]_1^3 = 3^2 - 1^2 = 9 - 1 = 8$. The required area is the value of the definite integral of the function $f(x) = 2x$ from $x = 1$ to $x = 3$. $\int_1^3 2x\,dx = [x^2]_1^3 = 3^2 - 1^2 = 9 - 1 = 8$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Properties of definite integrals**

■■ ■■ ■■■■■■■ ■■■■■■■■ ■■■■■■■■ ■■■■■■■ ■■■■■■■ $f(x) = 2x$ ■■ $x = 0$ ■■■ $x = 3$■ ■■■■■■ ■■■■■ ■■■■■■■■ ■■■ ■■■■■■■■■ What is the numerical value of the definite integral of the function $f(x) = 2x$ from $x = 0$ to $x = 3$, representing the area under the curve?

a. 6

b. **9** ✓

c. 12

d. 3

■■■■■■■ ■■■■■■■ ■■■■■■■■ ■■■■■■■■ ■■■■■■■■ ■■■■■■■ ■■■■■■■ $\int_a^b f(x)\,dx$. ■■ ■■■■ ■■■■■■■■ $f(x) = 2x$■ $a = 0$■ ■ $b = 3$. $\int_0^3 2x\,dx$ ■■■■■ ■■■■■■■■ ■■■■ ■■■■■■■■ ■■■■■■: $\int 2x\,dx = x^2 + C$. ■■■■■■ ■■■■■ ■■■■■■■■ ■■■■■■■■ ■■■■■■■■ ■■■■ ■■■■■■■■ ■■■■■■■■ ■■■■■■■■: $[x^2]_0^3 = (3)^2 - (0)^2 \ = 9 - 0 \ = 9$ ■■■■■■ ■■■■■■■■ ■■■■■■■■ ■■■■■■■■ ■■■■■■■■ ■■ 9. To find the numerical value of the definite integral, we use the formula $\int_a^b f(x)\,dx$. In this case, $f(x) = 2x$, $a = 0$, and $b = 3$..

30 **[Intermediate]** • **Target: Finding a function from its derivative**

■■ ■■ ■■■■■■■■ ■■■■ ■■■■■■■■ $f(x) = 2x$ ■■■■ $x = 0$ ■ $x = 3$■ What is the area under the curve $f(x) = 2x$ between $x = 0$ and $x = 3$?

a. 6

b. 9 ✓

c. 12

d. 3

■■■■■■ ■■■■■■■■ ■■■■ ■■■■■■■■ $f(x) = 2x$ ■■ $x = 0$ ■■■■ $x = 3$■ ■■■■■■■ ■■■■■■■■ ■■■■■■■: $\int_0^3 2x\,dx$ ■■■■■ ■■■■■■■ ■■■■■■■■ ■■ $2x$■ ■■■■ x^2. ■■ ■■■■■ ■■■■■■ ■■■■■■■■ ■■■■■■■■■: $[x^2]_0^3 = (3)^2 - (0)^2 = 9 - 0 = 9$ ■■■■■■ ■■■■■■■■ ■■ 9. To calculate the area under the curve $f(x) = 2x$ from $x = 0$ to $x = 3$, we use the definite integral: $\int_0^3 2x\,dx$ We find the antiderivative of $2x$, which is x^2. Then we apply the Fundamental Theorem of Calculus:
$[x^2]_0^3 = (3)^2 - (0)^2 = 9 - 0 = 9$ There.

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31 [Intermediate] • Target: Average value of a function

[illegible] $y = ax^2 + 1$ [illegible] [illegible] $x = 0$ [illegible] $x = 1$ [illegible] $\frac{7}{3}$[illegible] [illegible] [illegible] a. If the area of the region bounded by the curve $y = ax^2 + 1$, the x-axis, and the lines $x = 0$ and $x = 1$ is equal to $\frac{7}{3}$, find the value of the constant a.

a. 2

b. **3** ✓

c. 1

d. 4

[illegible]: $A = \int_0^1 (ax^2 + 1)\,dx$. [illegible] [illegible] [illegible]: $A = \left[\frac{ax^3}{3} + x\right]_0^1$. [illegible] [illegible]: $A = \left(\frac{a(1)^3}{3} + 1\right) - \left(\frac{a(0)^3}{3} + 0\right)$. [illegible] [illegible]: $A = \frac{a}{3} + 1$. [illegible] [illegible] [illegible] [illegible] $\frac{7}{3}$[illegible]: $\frac{a}{3} + 1 = \frac{7}{3}$. [illegible] 1 [illegible]: $\frac{a}{3} = \frac{7}{3} - 1 = \frac{7}{3} - \frac{3}{3} = \frac{4}{3}$. [illegible] [illegible] [illegible] 3: $a = 4$. The required ar.

32 [Intermediate] • Target: Word problems: integration in context

[illegible] [illegible] [illegible] [illegible] [illegible] $y = x^1$ [illegible] $x = 0$ [illegible] $x = 4$[illegible] Find the area under the curve $y = x^1$ from $x = 0$ to $x = 4$.

a. 4

b. 6

c. 5

d. **8** ✓

$\int_0^4 x^1\,dx = \frac{1}{2}x^2|_0^4 = 8$

33 [Intermediate] • Target: MCQ — Basic antiderivatives (the power rule)

[illegible] [illegible] [illegible] [illegible] $f(x) = 2x + 1$ [illegible] $x = 1$ [illegible] $x = 3$. Calculate the definite integral of the function $f(x) = 2x + 1$ from $x = 1$ to $x = 3$.

a. **8** ✓

b. 6

c. 10

d. 12

[illegible] [illegible] [illegible] [illegible] $f(x) = 2x + 1$ [illegible] $x = 1$ [illegible] $x = 3$[illegible] [illegible] [illegible] [illegible]: 1. [illegible] [illegible] [illegible] ([illegible] [illegible] [illegible]): $\int (2x + 1)\,dx = 2\int x\,dx + \int 1\,dx = 2\left(\frac{x^2}{2}\right) + x + C = x^2 + x + C$ [illegible] C [illegible] [illegible] [illegible]. 2. [illegible] [illegible] [illegible] [illegible] [illegible]: [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] (3) [illegible] [illegible] (1) [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]. $\int_1^3 (2x + 1)\,dx = [x^2 + x]_1^3$ 3. [illegible] [illegible] [illegible]: [illegible].

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34 [Intermediate] • Target: MCQ — Antiderivatives of polynomials

[illegible] [illegible] $f(x) = x^2 - 6x + k$ [illegible] [illegible] [illegible] [illegible] [illegible] $x = 3$. [illegible] [illegible] [illegible] k[illegible] The function $f(x) = x^2 - 6x + k$ represents a parabola that touches the x-axis at $x = 3$. What is the value of k?

a. **9** ✓

b. -9

c. 0

d. 3

[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] $x = 3$[illegible] [illegible] [illegible] [illegible] $x = 3$ [illegible] [illegible] [illegible] [illegible] [illegible] $f(x) = 0$. [illegible] [illegible] [illegible] $f(3) = 0$ [illegible] [illegible] [illegible] $\Delta = b^2 - 4ac$ [illegible] [illegible] [illegible] [illegible]. [illegible] [illegible] $x = 3$ [illegible] [illegible]: $f(3) = (3)^2 - 6(3) + k = 0$ $9 - 18 + k = 0$ $-9 + k = 0$ $k = 9$ [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] $x = 3$ [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]. [illegible] [illegible] [illegible] [illegible] (h, k) [illegible] [illegible] $ax^2 + bx + c$ [illegible] [illegible] $h = -\frac{b}{2a}$. [illegible] [illegible] [illegible] [illegible] $a = 1$ [illegible] $b = -6$. $h = -\frac{-6}{2(1)} = \frac{6}{2} = 3$ [illegible] [illegible] [illegible] [illegible] [illegible] [illegible].

35 [Intermediate] • Target: MCQ — Antiderivatives of trigonometric and exponential functions

[illegible] [illegible] [illegible] [illegible] [illegible] [illegible]. [illegible] [illegible] [illegible] $f(t) = -5t^2 + 20t + 1$ [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]

36 [Intermediate] • Target: MCQ — Evaluating definite integrals

[illegible] $f(x) = a \cdot e^{kx}$[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] $(0, 5)$ [illegible] $(\ln(3), 15)$. [illegible] [illegible] [illegible] a [illegible] [illegible] k. For the exponential function $f(x) = a \cdot e^{kx}$, two points on its graph are $(0, 5)$ and $(\ln(3), 15)$. Find the value of the constant a and the value of the constant k.

a. $a = 5, k = \ln(3)$

b. $a = 5, k = 3$

c. $a = 5, k = \frac{\ln(3)}{1}$

d. $a = 5, k = \ln(15/5)$ ✓

[illegible] [illegible] [illegible] $(0, 5)$ [illegible] [illegible] [illegible] [illegible] $f(x) = a \cdot e^{kx}$[illegible] [illegible] [illegible] $x = 0$ [illegible] $f(x) = 5$: $5 = a \cdot e^{k \cdot 0}$ $5 = a \cdot e^{0}$ $5 = a \cdot 1$ $a = 5$ [illegible] [illegible] [illegible] $a = 5$. [illegible] [illegible] [illegible] $(\ln(3), 15)$ [illegible] $a = 5$ [illegible] k: $15 = 5 \cdot e^{k \cdot \ln(3)}$ [illegible] [illegible] [illegible] 5: $3 = e^{k\ln(3)}$ [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]: $\ln(3) = \ln(e^{k\ln(3)})$ $\ln(3) = k\ln(3)$ [illegible] [illegible] [illegible] $\ln(3)$ ([illegible] [illegible] [illegible] $\ln(3) eq 0$): $k = 1$ [illegible] $a = 5$ [illegible] $k = 1$. Since.

37 [Intermediate] • Target: MCQ — Definite integrals with polynomial integrands

[illegible] [illegible] $f(x) = 3x^2 + 2x - 1$[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] $\int_0^2 f(x)\,dx$[illegible] If $f(x) = 3x^2 + 2x - 1$, what is the value of the definite integral $\int_0^2 f(x)\,dx$?

a. 14 ✓

b. 10

c. 12

d. 8

[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] $f(x)$[illegible] [illegible] [illegible] [illegible] [illegible] [illegible]. [illegible] [illegible] [illegible] [illegible] $f(x) = 3x^2 + 2x - 1$ [illegible] $\int (3x^2 + 2x - 1)\,dx = x^3 + x^2 - x + C$. [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] 0 [illegible] 2: $\int_0^2 (3x^2 + 2x - 1)\,dx = [x^3 + x^2 - x]_0^2 = (2^3 + 2^2 - 2) - (0^3 + 0^2 - 0) = (8 + 4 - 2) - (0) = 10$. To find the value of the definite integral, we first find the antiderivative of the function $f(x)$, and then substitute the limits of integration. The anti.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: MCQ — Definite integrals of trigonometric and exponential functions • Cross-Lesson: 8.01 Exponentials & Logarithms

[illegible] [illegible] $f(x) = ax^2 + bx + c$ [illegible] [illegible] [illegible] [illegible] $(2, -5)$ [illegible] [illegible] $(4, 7)$. [illegible] [illegible] [illegible] a[illegible] A quadratic function $f(x) = ax^2 + bx + c$ has its vertex at $(2, -5)$ and passes through the point $(4, 7)$. What is the value of a?

a. 2

b. **3** ✓

c. 4

d. 1

[illegible] [illegible] [illegible] (h, k) [illegible] [illegible] [illegible] $f(x) = a(x-h)^2 + k$. [illegible] [illegible] [illegible] $h = 2$ [illegible] $k = -5$[illegible] [illegible] $f(x) = a(x-2)^2 - 5$. The quadratic function with vertex (h, k) can be written in the form $f(x) = a(x-h)^2 + k$. In this case, $h = 2$ and $k = -5$, so $f(x) = a(x-2)^2 - 5$. [illegible] [illegible] [illegible] [illegible] [illegible] $(4, 7)$[illegible] [illegible] [illegible] [illegible] [illegible] $x = 4$ [illegible] $f(x) = 7$ [illegible] a. We know the function passes through the point $(4, 7)$, so we can substitute $x = 4$ and $f(x) = 7$ to find a. $7 = a(4-2)^2 - 5$.

39 [Challenging] • Target: MCQ — Area under a curve • Cross-Lesson: 9.01 Statistics & Probability

[illegible] [illegible] [illegible] [illegible]: $\int_{-2}^{2}(x^3 + x)\,dx$[illegible] Integral of odd function on symmetric interval?

a. **0** ✓

b. 4

c. 8

d. -4

[illegible] $f(x) = x^3 + x$ [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] $[-a, a]$ [illegible] [illegible] [illegible].

40 [Challenging] • Target: MCQ — Signed area and regions below the axis • Cross-Lesson: 1.01 Algebra & Equations

[illegible] [illegible] [illegible] [illegible]: $\int_{a}^{a} f(x)\,dx$ [illegible] [illegible] $f(x)$[illegible] Definite integral with identical upper and lower limits?

a. **0** ✓

b. 1

c. $f(a)$

d. $2a$

[illegible] [illegible] [illegible] [illegible]: [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]: $\int_{a}^{a} f(x)dx = 0$.

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41 [Challenging] • Target: MCQ — Properties of definite integrals • Cross-Lesson: 2.01 Functions & Transformations

■■ ■■■■ ■■■■■■■ ■■■■■■■: $\int_1^2 \frac{2x+1}{x^2+x}\,dx$■ Evaluate integral where numerator is derivative of denominator.

a. $\ln 3$ ✓

b. $\ln 6$

c. $\ln 2$

d. 1

$[\ln(x^2+x)]_1^2 = \ln(4+2) - \ln(1+1) = \ln 6 - \ln 2 = \ln\left(\frac{6}{2}\right) = \ln 3$.

42 [Challenging] • Target: MCQ — Finding a function from its derivative • Cross-Lesson: 3.01 Sequences & Series

■■ ■■■■ ■■■■■■■: $\int \frac{2x}{x^2+1}\,dx$■ Evaluate integral where numerator is derivative of denominator.

a. $\ln(x^2+1)+C$ ✓

b. $\frac{1}{x^2+1}+C$

c. $\arctan(x)+C$

d. $2\ln(x^2+1)+C$

■■■ ■■ ■■■■■ ■■ ■■■■■ ■■■■■■ ■■■■■■■ ■■■ ■■■■■■■ ■■■■■ ■■■■■■■■■■ ■■■■■■■ ■■■■■■: $\ln(x^2+1)+C$.

43 [Challenging] • Target: MCQ — Average value of a function • Cross-Lesson: 4.01 Trigonometry

■■■ ■■■■ $f(x)$ ■■■■ ■■■■■ ■■■■ $\int_0^3 f(x)\,dx = 5$■ ■■■ ■■■■ $\int_{-3}^3 f(x)\,dx$■ Integral of even function on symmetric interval?

a. **10** ✓

b. 0

c. 5

d. -5

■■■■■ ■■■■■■ ■■■■■■■ ■■■ ■■■■ ■■■■■■■ ■■■■■ ■■■ ■■■■■■■ ■■■ ■■■ ■■■■■■: $2 \times 5 = 10$.

44 [Challenging] • Target: MCQ — Word problems: integration in context • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

■■■ ■■■ $\int_1^4 f(x)\,dx = 8$■ ■■■ ■■■■ $\int_1^4 3f(x)\,dx$■ Constant multiple property of integrals?

a. **24** ✓

b. 8

c. 11

d. 12

■■■■ ■■■■■■ 3 ■■■■ ■■■■■■■: $3\int_1^4 f(x)dx = 3(8) = 24$.

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45 [Challenging] • Target: Basic antiderivatives (the power rule) • Cross-Lesson: 6.01 Limits & Derivatives

■■ ■■■■ ■■■■■■■ ■■■■■■: $\int_0^3 2x\,dx$■ Evaluate definite integral $\int_0^3 2x dx$.

a. **9** ✓

b. 6

c. 3

d. 18

$[x^2]_0^3 = 3^2 - 0 = 9$.

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